

Rajya Stariya Bal Vaigyanik Pradarshani 2023–24

Theme : Science and Technology for Society
Subtheme: LiFE (Lifestyle for Environment)

Transforming Cremations with Sustainable Innovations



CowDung- Coconut Waste Eco-Logs to Replace Tree-wood

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Delhi Public School Chandigarh

Segregating Waste at Source – The Way to Go

Khushi Sharma



DO we pause and think even once before tossing a plastic bottle in the garbage or throwing a peel of banana in the same bin? How and where does the waste we produce go?

We hear of plastic waste forming islands in oceans and deeply impairing marine life. Piled up garbage slowly gets covered by the soil and forms an impermeable layer in our earth's crust thwarting the natural process of water infiltration.

We all want our houses and our schools to be clean. We throw what is called 'the solid waste' out of our premises. Yet when we see piles of garbage lying on the roadside or garbage bins overflowing in our vicinity we simply try to pass on the buck to the local civic bodies. Our social perceptions have been formed on these lines. The city should appear clean but is it actually cleaned up? Where is the waste going, isn't that also our business? Do we have a role to play?

I never realised this till I got an opportunity to work hands-on with the Mohali Municipal Corporation (MC) in Punjab as part of the SEWA project, as required under the CBSE curriculum for Class X.

To begin with, let us understand the problem of 'Solid Waste'. Interestingly Solid Waste in itself is not a problem rather it is an asset. It has value if properly managed. The problem is poor systems of Solid Waste Management (SWM) and this is the bane of our society.

Today SWM not only consumes huge budgets of municipal bodies, it incurs high social costs too. In the Mohali MC, a large budget goes towards SWM – mostly towards salaries of staff. Then there are adverse health impacts on waste collectors, rag pickers, communities living near garbage bins and dumpsites. With economic growth, change in lifestyles and food habits our per capita waste generation has increased. The Mohali MC

produces 82.5 Tonnes of solid waste per day. So we need a large workforce for waste collection, transportation, processing and then landfilling.

Chandigarh, the State capital and a planned city with one million population, has over 45 acres of land earmarked for landfill site but most cities in India are overpopulated and do not even have adequate land availability. So, invariably the vacant lands near the residential areas turn into huge trash bins. Even here in Chandigarh, the dumping site has become a huge mountain of garbage. Indiscriminate dumping of solid waste in landfill not only leads to ground water contamination, land and air pollution, but also requires more land. Land near cities is costly while land far from cities involves high costs for waste transportation.

For effective SWM, the collection capacities should be greater than or equal to the waste

generation rates. However our municipalities are often short-handed.

Besides, the waste collector's or rag picker's job is not something they would cherish. Think about it – would I like to soil my hands in my own household dustbin? No. How then do I expect some other human being to play around with my unhygienic waste. No wonder the waste collectors are ill motivated, irregular and have low productivity. As per the India Water Portal, the collection efficiency of municipal solid waste ranges from 22% to 60%.

The simple solution to manage waste is not just technology but also change in our culture. We need to practice waste segregation in our homes, then the whole city, state and country's waste management can be effectively managed with little technological intervention.

What is required from each waste generator is to keep separate the kitchen and garden waste which in popular parlance is called the wet waste from the remaining dry waste. Wet waste has high organic content. Composting is the most preferred and appropriate way to convert this waste into useful organic manure. Wet waste forms over 50% of the municipal waste. Composting can be done in households, institutions and in local public parks. This considerably reduces the quantum of waste to be transported over long distances.

The rest of the waste which has plastic, glass, cardboard, paper, etc. acquires a value when not spoilt by the wet waste. Waste collectors are happy to collect them, recover the various materials and sell them to the chain of *kabbadis* – an informal system that promotes recycling and re-use of materials such as the packaging paper bags used by roadside hawkers.

Mohali is a rapidly developing IT hub and an important commercial city of the State of Punjab, bordering the capital

Pupils ask residents to segregate waste



Students during their field trip organised by the Mohali civic body

MOHALI, JULY 2

A field trip was organised by the Municipal Corporation for students of Delhi Public School (DPS) under a door-to-door waste collection and segregation campaign.

Suhani Sharma from the DPS, Chandigarh, led a team of around 10 students during the field trip. They visited residences and motivated people to

segregate waste into dry, wet and hazardous trash before handing it over to the collector.

They watched operations at the material recovery facility of the resource management centre, Phase 3-A; gaushala construction and demolition waste management unit in the Industrial Area and horticulture waste management unit at the nature park. — TNS

Cremation of the dead



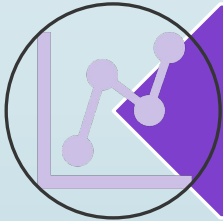
Hindus, Sikhs, Jains, Buddhists cremate their dead.
Conventionally, Firewood is used for cremations



Mortality Rate 2022 ~ 9 deaths per 1000 persons
Estimated cremations in India, 2023 ~10 million

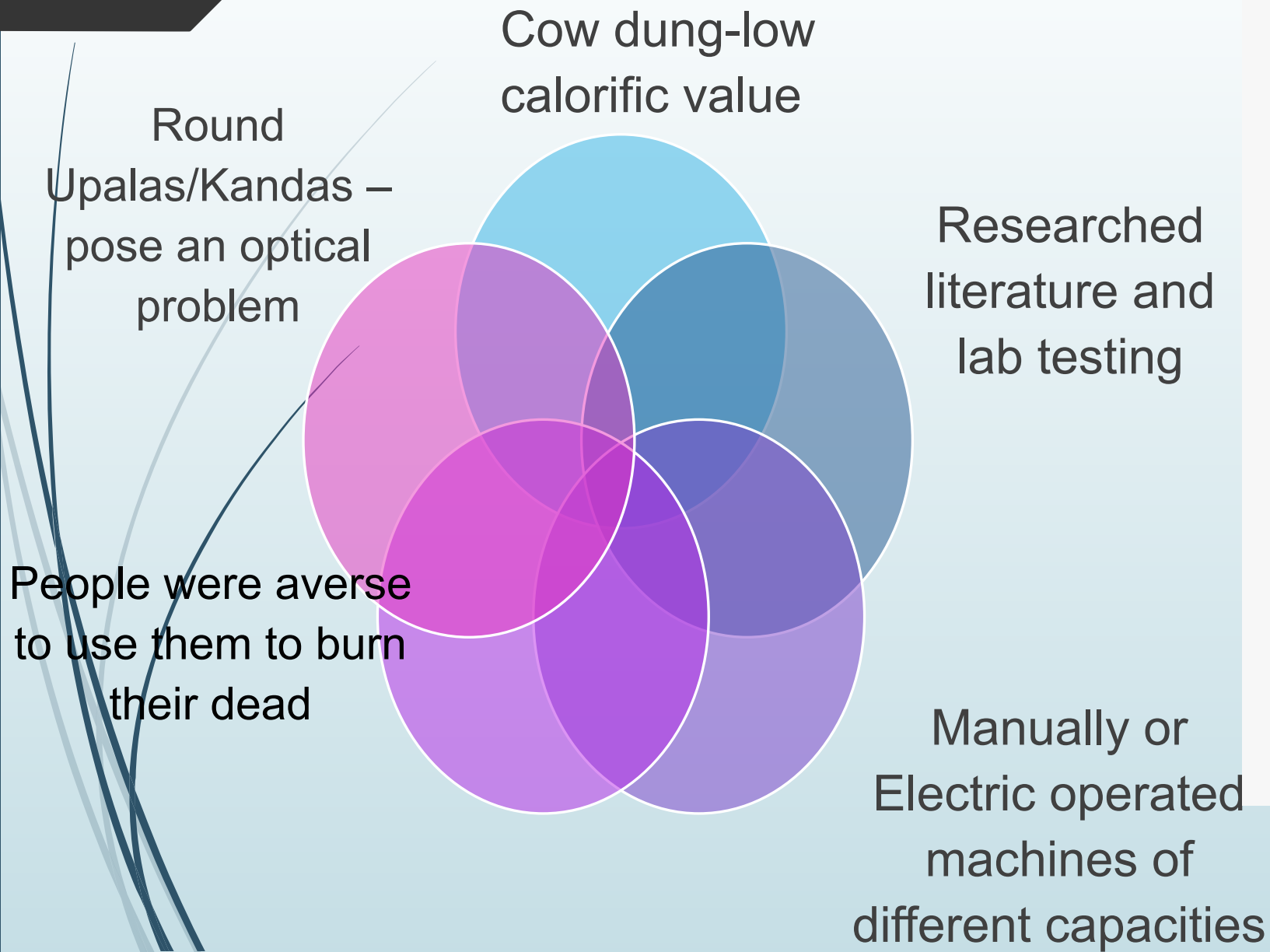


High Demand of Wood-Mango, Neem, Banyan,
Eucalyptus trees with potential life 100s of years



One Average cremation needs
~ 500 kg of wood = 2 trees grown over 10-15 years

Cow dung waste shaped into wood logs



Wasted Green Coconut Shells at RMC, Mohali



- ❑ Increased consumption of coconut water and pulp in North India has increased generation of coconut shell waste.
- ❑ About 1.5 kg of shell is lost from each coconut consumed
- ❑ Estimated waste in the country-Three million tons per year
- ❑ Coconut shells can take up to ten years to degrade in the environment
- ❑ Causes unsightliness in urban environment
- ❑ **Breeds mosquitoes in left over water** leading to disease transmission

Convinced MC Mohali for Shredding of Coconut Shell waste



At Resource Management Centre
Sorting and Processing Municipal Waste, MC Mohali

Scientific Thought into Social Action

Feed cow dung and shredded coconut shells into the Hooper

Machine mixes, shapes and extrudes out long logs

Different molds make different log sizes

A Cylindrical hole in the center facilitates easy drying & combustion

Sun-dried logs – hard and sturdy

Log Length: 2.5-3 ft and diameter can vary 3" to 6"

Log weight(wet): 3-4kg, Dry log weighs 1kg





A model Set-up

Making Eco-Logs
At Mohali
Gaushala, for
supply to Balongi
Crematorium

Innovation: Replacing conventional crematorium wood with Ecologs crafted from a blend of cowdung and coconut shell waste

- ❑ Cow-Dung waste chokes sewers and causes water pollution. This innovation creates demand and value for even the non-milk-producing cows
- ❑ Shredding coconut shells checks spread of Dengue, Malaria

While saving trees, this innovation solves waste management and public health concerns



Sun-Drying of Eco-Logs

Testing and Innovation

Testing done at Thapar Technology Centre, Patiala

Calorific Value of Wood ~ 3500 kcal/kg

Calorific value of dried pure cow-dung
~1640 kcal/kg

Adding 1-part shredded coconut shells to
5 parts cow-dung 3300 kcal/kg

Improved Log Quality– Strength, Quick Dryness

Testing Results from the Lab: With addition of 1 part of coconut peat to 5 parts of cow-dung; its calorific value got equivalent to a wood powder ~ 3500 kcal/kg (calorific value of firewood is lower than (80%) of wood powder due to moisture)

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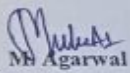
TEST REPORT

Sample ID	NN/23-24/404	Date:	27.09.2023
Service No.	NN/23-24/404 (01)	Customer's Ref.	Sample submitted by customer on dtd 25.09.2023
Customer's name and address:			
M/s Punjab Municipal Infrastructure Development Company 5 - Punjab Municipal Bhawan Dakshin Marg, 35 A, Sector 35 A Mohali (Pb) Kind Attn. : Mr. Dalip Kumar			
Sample Description	Wood Powder		
Condition of the sample received	O.K.		
Customer's sample identification No. (if any)	—		
Quantity/number of samples	500 gm / 1		
Sampling Procedure (if any)	—		
Test parameters	GCV		
Standard/Specification/Method followed	As mentioned below		
Deviations (if any)	—		
Documents constituting this report (if any)			
Date of Receipt of Job		Date of Completion of Job	Total Number of Pages
25.09.2023		27.09.2023	1

TEST RESULTS

Sr. No.	Parameters	Test Method	Unit	Results (As received basis)
1	Gross Calorific Value (GCV) (Oven Dried basis)	IS: 1350 (Part 2)- 2022, (2 nd revision) Isothermal	Kcal/kg	3597

.....End of test report.....


M. Agarwal
Technical Manager
(Authorized Signatory)

Note: 1. The results listed refer only to the tested samples and applicable parameters. Endorsement of products is neither inferred nor implied.
2. Samples will be destroyed after one month (except water, wastewater) from the date of issue of the test report unless otherwise specified.
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4. In case any reconfirmation of contents of the test report is required, please contact the authorized signatory of the test report within 15 days of the issue of test report.

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URL: www.sailabs.org SAIL/FM/CSC-1

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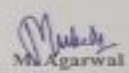
TEST REPORT

Sample ID	NN/23-24/384	Date:	22.09.2023
Service No.	NN/23-24/384 (01)	Customer's Ref.	Sample submitted by customer on dtd 19.09.2023
Customer's name and address:			
M/s Punjab Municipal Infrastructure Development Company 5 - Punjab Municipal Bhawan Dakshin Marg, 35 A, Sector 35 A Mohali (Pb) Kind Attn. : Mr. Dalip Kumar			
Sample Description	Mixture of Cow Dung & Coco Peat		
Condition of the sample received	O.K.		
Customer's sample identification No. (if any)	—		
Quantity/number of samples	150 gm / 1		
Sampling Procedure (if any)	—		
Test parameters	GCV, Moisture		
Standard/Specification/Method followed	As mentioned below		
Deviations (if any)	—		
Documents constituting this report (if any)			
Date of Receipt of Job		Date of Completion of Job	Total Number of Pages
19.09.2023		21.09.2023	1

TEST RESULTS

Sr. No.	Parameters	Test Method	Unit	Results
1	Gross Calorific Value (GCV) (Oven Dried basis)	IS: 1350 (Part 2)- 2022, (2 nd revision) Isothermal	Kcal/kg	3371
2	Moisture @ 108 °C	IS: 1350 (Part 1)-2004 (2nd revision) Reaffirmed 2019 Gravimetric	%	82.05

.....End of test report.....


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Scalable, Replicable, Sustainable

No dearth of Basic Raw material – Cow-dung

Cocopeat can be complemented with horticulture waste like Dry Leaves or fallen tree branches

Lab Testing reveals that adding dry leaves in the same proportion as shredded green coconut has equivalent calorific value > 3000 kcal/kg



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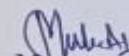
TEST REPORT

Sample ID	NN/23-24/404	Date:	27.09.2023
Service No.	NN/23-24/404 (02)	Customer's Ref.	Sample submitted by customer on dtd 25.09.2023
Customer's name and address:			
M/s Punjab Municipal Infrastructure Development Company			
5 - Punjab Municipal Bhawan			
Dakshin Marg, 35 A, Sector 35 A			
Mohali (Pb)			
Kind Attn. : Mr. Dalip Kumar			
Sample Description		Cow Dung + Dry Leaves	
Condition of the sample received		O.K.	
Customer's sample identification No. (if any)		--	
Quantity/number of samples		50 gm / 1	
Sampling Procedure (if any)		--	
Test parameters		GCV	
Standard/Specification/Method followed		As mentioned below	
Deviations (if any)		--	
Documents constituting this report (if any)		--	
Date of Receipt of Job		Date of Completion of Job	Total Number of Pages
25.09.2023		27.09.2023	1

TEST RESULTS

Sr. No.	Parameters	Test Method	Unit	Results (As received basis)
1	Gross Calorific Value (GCV) (Oven Dried basis)	IS: 1350 (Part 2)- 2022, (2 nd revision) Isothermal	Kcal/kg	3129

.....End of test report.....


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Sustainable Life-style Innovation

- ☐ Blended Logs are stronger, Not brittle like the cow-dung logs
- ☐ Public Acceptability is High due to use of cow dung
- ☐ Easy to Handle and Transport
- ☐ Easy and Quick to Dry
- ☐ Cocopeat improves Quality
- ☐ Emissions are lower than Firewood
- ☐ Cost-Effective



Scientific thought translated into Social Action

Truckloads of Logs on way to the Crematorium



Maths : Eco-Friendly Lifestyle Saving Trees

- 5 Tons of Cow-dung Daily (~600 cows, Each cow produces ~0.8 kg dung)
+
0.75 ton of Coconut peat (~500 shells, Each shell ~ 1.5 kg waste)
= 2 Tonnes of Logs (Cow Dung Dries up to 25% Weight)
Equivalent to 2 Tonnes of Firewood used for Cremation
- (Lab Test shows equivalent calorific value of Eco-logs and firewood)
= 4* 500 kg wood used per cremation
- Average 8 cremations happen at Balongi crematorium each day
- Saves 50% of Tree wood = Saves 8 Trees everyday (500 kg wood is Equivalent to 2 trees)

Cost Effective Eco-Friendly Sustainable Solution

- Locally, Tree wood costs Rs 8 per kg
 - 500 kg of tree-wood needed per cremation costs Rs 4000/-
 - Supply Cost of Blended Eco-Logs ~ 3/- per kg
(Incl. Labour, Space for Shed, Drying, Transport*etc)
- *The Crematorium is 1.5 Km away from the Gaushala
- With time, cost of Tree-wood is likely to appreciate as Forests are declining

The Eco-Logs costs one-third of the Tree-wood

Demonstrating other Cow-dung based Products

Industrial Area, Sector 73 Mohali Gaushala produces ~ 6 tons of cow-dung daily. Besides, cow-dung logs, we also make-

Vermi-compost (Vermicompost boosts the nutrients available to plants, helping seeds to germinate more quickly, grow faster, develop better root systems and produce higher yields.)

Flower pots for growing seedlings in the nurseries rather than plastic bags. (The cow-dung pots give nutrition to the saplings and can be planted in the soil along with their plants.)

Diyas for lighting lamps for festivals – can be disposed off in the soil, adding to soil fertility.

